

Seonghyun Jin

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Research Interests

Reconstructing and generating the real 3D world; streaming 3D reconstruction; geometric perception; latent state modeling; generative world modeling.

Education

KAIST, Graduate School of AI

Mar. 2025 – Present
Daejeon, Korea

Integrated M.S./Ph.D. student in Artificial Intelligence

- BISPL Lab, advised by [Prof. Jong Chul Ye](#).

Seoul National University

Mar. 2020 – Feb. 2025
Seoul, Korea

B.S. in Computer Science; B.A. in Economics, College of Liberal Studies

- GPA: 3.68 / 4.3. Graduated *Cum Laude*.

Publications

FILT3R: Latent State Adaptive Kalman Filter for Streaming 3D Reconstruction

Seonghyun Jin, Jong Chul Ye

Under review. [arXiv:2603.18493](#).

[Paper](#) [Code](#) [Project](#)

CRePE: Curved Ray Expectation Positional Encoding for Unified-Camera-Controlled Video Generation

Seonghyun Jin, Youngmin Kim, Sunwoo Park, Jong Chul Ye

Under review. Preprint in preparation, 2026

[arXiv link](#) to be added after upload.

AI-Based Mental Health Assessment for Adolescents Using Their Daily Digital Activities

Do Hyung Kim, Joonsung Lee, Taehwi Lee, Soeun Baek, **Seonghyun Jin**, HaEun Yoo, Youngeun Cho, Seonghyeon Park, Kwangsu Cho, Chang-Gun Lee

IEEE DSAA, 2024

[Paper](#) 6 citations

Previous Research Experience

RUBIS Lab, Seoul National University

Sep. 2022 – Jun. 2024
Seoul, Korea

Undergraduate Research Intern

- Researched digital phenotyping for adolescent mental health assessment from daily digital activity data at [RUBIS Lab](#).
- Performed feature engineering on a human-stroke/scribble dataset for downstream AI-based mental health prediction.
- Built a React-based data collection application supporting a longitudinal user study with ~300 adolescent participants over 2 months.
- Contributed to publication at IEEE DSAA 2024 (co-author).

Languages

Korean Native

English Fluent; TEPS 538, TOEIC 980

Technical Skills

Programming	Python, C, Dart, PyTorch, Git, L ^A T _E X
Research	3D reconstruction, computer vision, deep learning, experimental evaluation